



Information Systems and Digital Inclusion Laboratory

LUDOLAB

E-INCLUSION PROJECT

VIRTUAL AND FACE-
TO-FACE MODALITY

USER MANUAL

Excel

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Author

Jean Pierre Cotera Alcivar

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1. Project Objectives

To train Ministry of Education teachers in the use of ICT tools to improve their teaching performance, in order to promote educational innovation and online learning.

Train priority groups or digital illiterates in the use of ICT tools to apply what they have learned in their daily lives, promoting their digital inclusion and improving their quality of life.

2. Objective of the Manual

The fundamental objective of this manual is to provide a detailed and accessible guide to train priority groups or digitally illiterate individuals in the use of information and communication technologies (ICT), with a specific focus on the practical application of Microsoft Excel. We seek to empower these individuals by providing them with the necessary skills to effectively integrate this spreadsheet tool into their daily lives. Through this training process, we not only aim to improve the participants' digital competence, but also to foster their inclusion in today's society, where the use of tools such as Excel can open doors to job opportunities and improvements in various aspects of their lives. This manual is conceived as a key tool to improve the quality of life of these communities, allowing them to take full advantage of the benefits that Excel and other digital technologies can offer.

3. Scope

The scope of this tutorial covers several essential functions of Microsoft Excel, focusing specifically on the training of priority groups or digital illiterates. Fundamental topics for the effective management of this spreadsheet tool will be addressed, including adding cells to perform mathematical operations, inserting columns and rows to organize and structure data, as well as applying colors to highlight relevant information and improve the readability of spreadsheets.

This tutorial aims to provide clear, step-by-step instructions that facilitate the understanding and practical application of these functions in everyday situations. By addressing these specific capabilities, it seeks to equip participants with the necessary skills to use Excel effectively in their daily lives, providing them with concrete tools that they can apply in a variety of areas, from managing personal budgets to organizing data relevant to their daily activities.

4. Aimed at

This tutorial is aimed especially at priority target groups, including the elderly and those who are digitally illiterate. The main focus is to offer training tailored to the specific needs of these participants, recognizing the possible barriers they may face in the use of information and communication technologies (ICT).

By addressing this target audience, the tutorial seeks to be inclusive and accessible, providing a learning space designed to address the particularities and challenges that may arise in the process of familiarization with Microsoft Excel. Through clear instructions and practical examples, the aim is to provide these people with the necessary tools to overcome any initial reluctance and make the most of Excel's functionalities in their day-to-day work.

5. Application introduction

Microsoft Excel is a widely used spreadsheet application that provides users with the ability to organize, analyze and present data efficiently. Its fundamental concept lies in providing a versatile platform for creating and manipulating electronic spreadsheets. With Excel, users can perform a variety of functions, including:

- a) **Cell Sum:** Excel allows you to quickly add values of selected cells, making it easier to perform mathematical operations and obtain totals.
- b) **Insertion of Columns and Rows:** The application allows adding or deleting columns and rows as needed, enabling the efficient organization and structuring of information.
- c) **Color Placement:** Excel provides the option to apply visual formatting through the use of colors, which facilitates quick identification of important data and improves the overall presentation of the spreadsheet.
- d) **Charting:** Excel allows you to effectively visualize data through the creation of charts. From simple bar charts to complex scatter plots, users can visually represent their data to gain a clearer and faster understanding of trends and patterns.
- e) **Use of Formulas:** Excel is known for its robust formula system, which allows you to perform complex calculations and data analysis in an automated manner. Predefined functions, such as SUM, AVERAGE and VLOOKUP, facilitate data manipulation and processing, saving time and reducing potential errors.
- f) **Power BI Integration:** Integration with Power BI, another Microsoft tool, extends Excel's analytical capabilities. Power BI enables the creation of interactive dashboards and dynamic reports, taking data analysis to a higher level and facilitating informed decision making.

These essential functions are just a few of the many capabilities offered by Excel. The application stands out for its flexibility and usefulness in a wide range of contexts, from managing budgets and tracking expenses to creating reports and charts for more visual presentations. In short, Microsoft Excel is a powerful tool that provides users with the ability to work efficiently with numerical data and improve the organization and understanding of information.

6. Basic Notions

Before diving into the use of Microsoft Excel, it is essential that the user acquires a basic understanding of the application's working environment and its component elements. The following is a description to guide the user:

Microsoft Excel Workspace:

- a) **Excel workbook:** The main file is called a "workbook" and contains individual spreadsheets. Each sheet is referred to as a "spreadsheet" and is where data is entered and organized.^{[1][1]}
- b) **Cells:** The Excel interface is organized in cells, which are tables arranged in rows and columns. Each cell is identified by a letter and a number (e.g., A1, B2) and can contain data, formulas or functions.^[1] The cells are organized in cells, which are tables arranged in rows and columns.
- c) **Formula Bar:** This is located at the top of the Excel window and displays the formula or the contents of the active cell. This is where you can enter data directly or perform mathematical operations.
- d) **Quick Access Toolbar:** Located at the top left, it allows quick access to common functions such as Save, Undo and Redo.
- e) **Options Ribbon:** The ribbon, located at the top of the window, groups functions into tabs such as "Home", "Insert", and "Format", making it easy to navigate and access various tools.^[1] The ribbon is located at the top of the window.

Basic Elements and Functions:

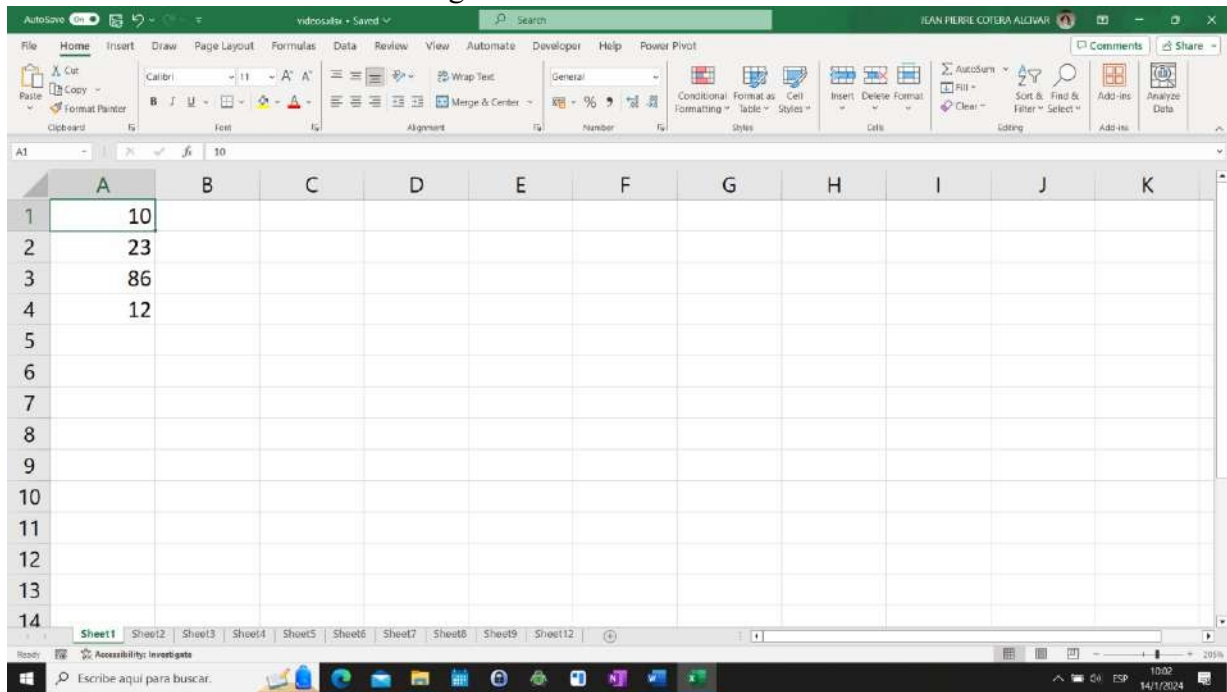
- a) **AutoSum:** To quickly perform sums, select the cell where you want to see the result, click the AutoSum button on the toolbar or use the keyboard shortcut (Alt + =). Excel will automatically identify the nearby range of cells and generate the sum.
- b) **Cell Coloring by Rules:** You can highlight specific data using conditional formatting rules. Select the range of cells, go to "Conditional Formatting" in the Home tab, and choose a rule, such as highlighting cells greater than or less than a given value.
- c) **Inserting Columns and Rows:** To insert columns or rows, select the column or row adjacent to the desired location, right click and select "Insert". This will move the existing content and provide space for new columns or rows.

Understanding these basic workspace elements and essential functions will make it easier for users to learn and take advantage of Microsoft Excel's capabilities.

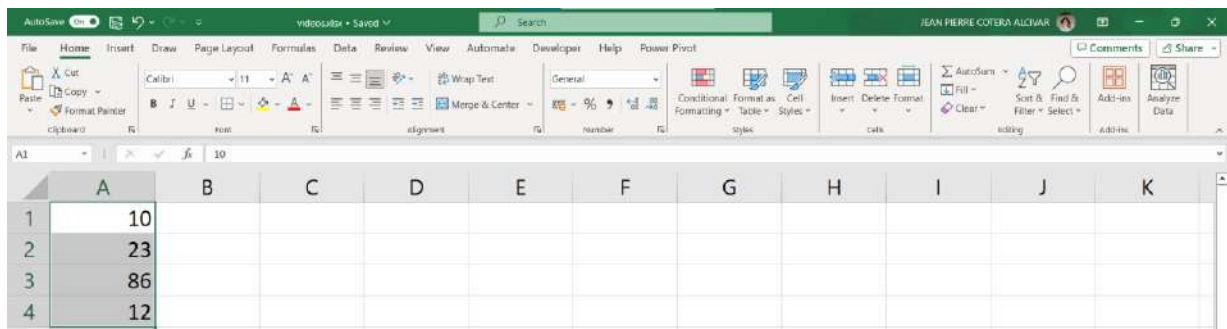
7. Steps to follow

Sum of cells

a. First We have the following data that we want to sum.



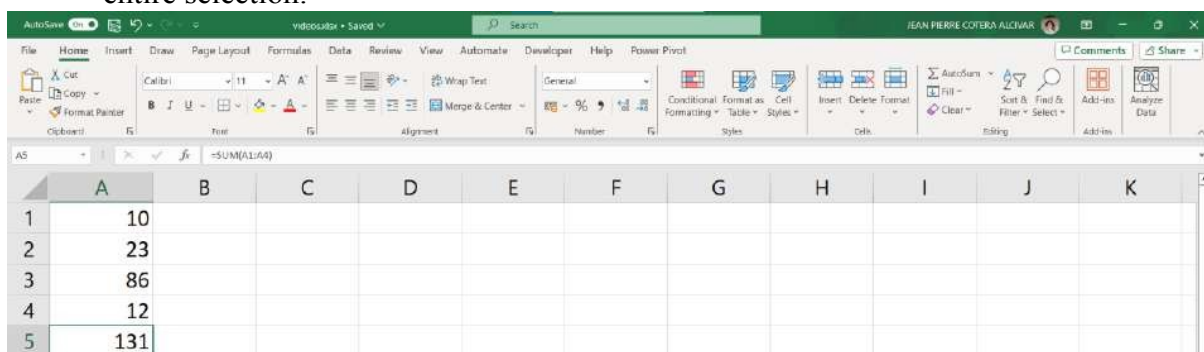
b. Now we select the cells we want to add



c. Once this is done, we go to the home tab and the AutoSum function

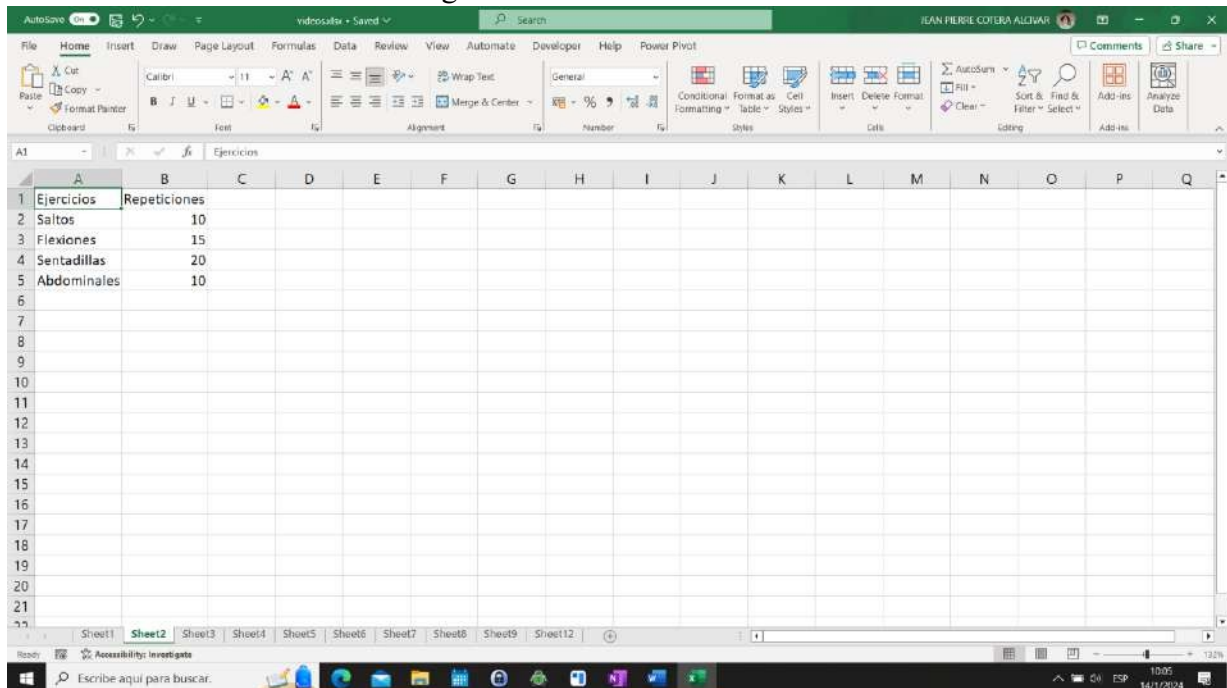


d. Once you click on auto sum, the result of the sum will be displayed one cell below the entire selection.



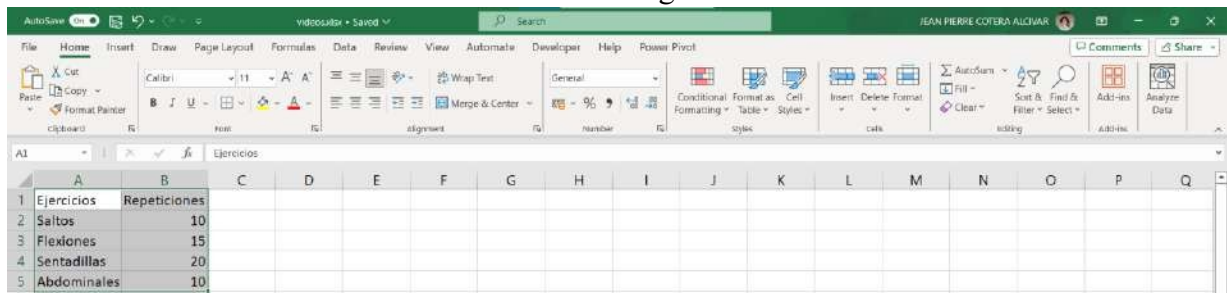
Insert a graphic

a. First we have the following data in our Excel



Ejercicios	Repeticiones
Saltos	10
Flexiones	15
Sentadillas	20
Abdominales	10

b. Now we have to select all the data including their headers



Ejercicios	Repeticiones
Saltos	10
Flexiones	15
Sentadillas	20
Abdominales	10

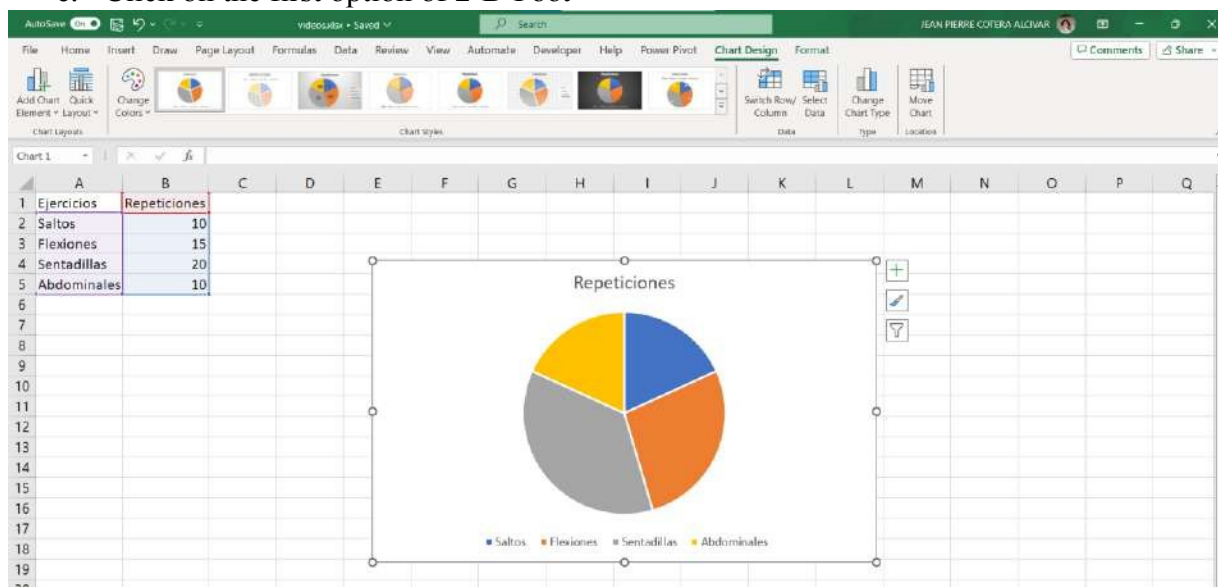
c. Once this is done, we go to the insert tab and then choose a graphic.



d. Click on the circle and the following window will appear



e. Click on the first option of 2-D Foot



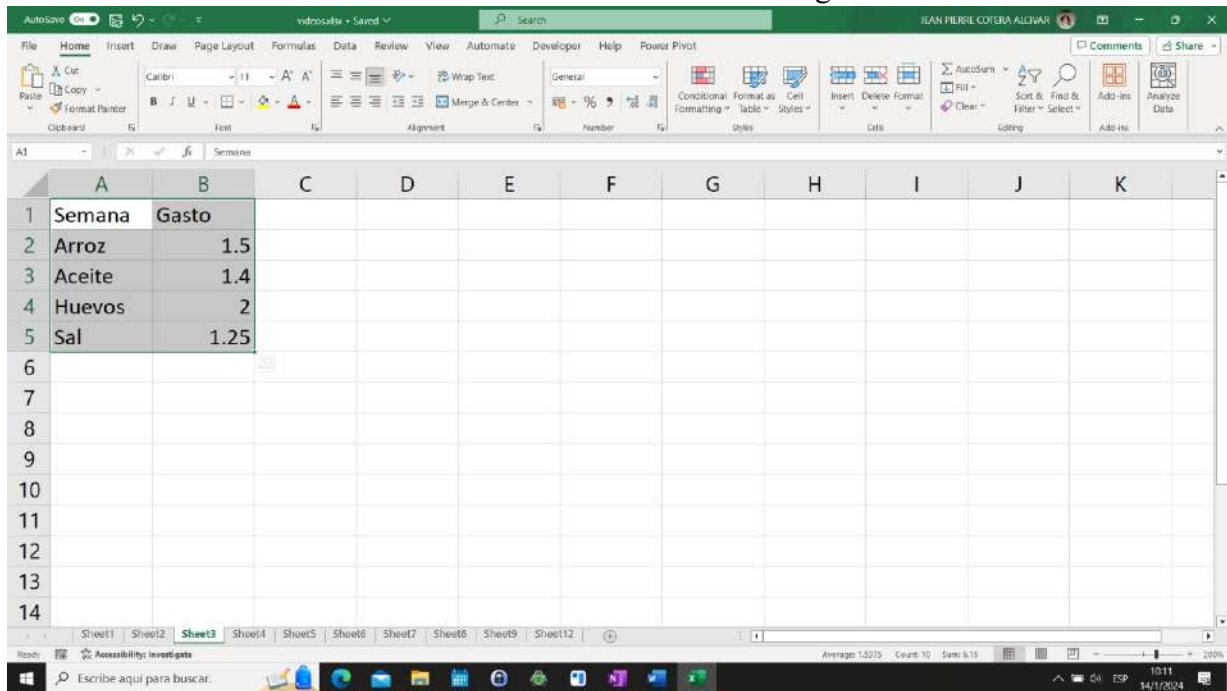
Customize a table

a. We have the following data

Semana	Gasto
Arroz	1.5
Aceite	1.4
Huevos	2
Sal	1.25

The screenshot shows the Excel interface with the 'Home' tab selected. The table is located in the range A1:B5. The status bar at the bottom indicates the active sheet is 'Sheet3'.

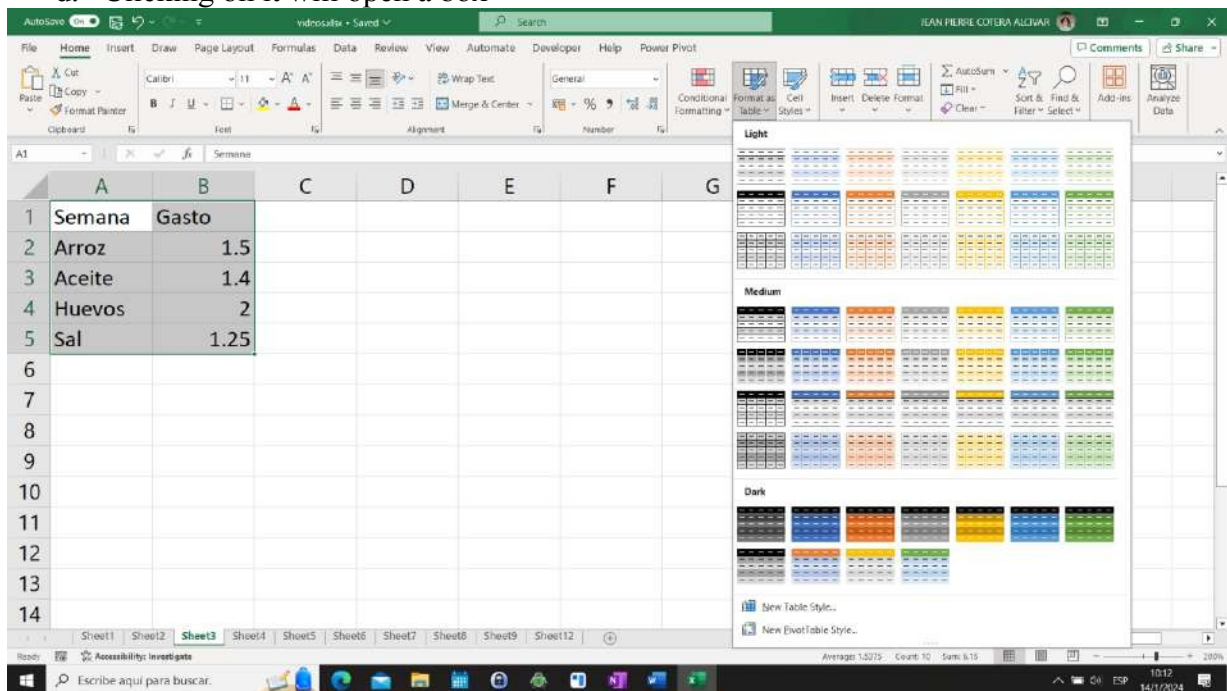
b. To customize it we must select the entire table including its headers.



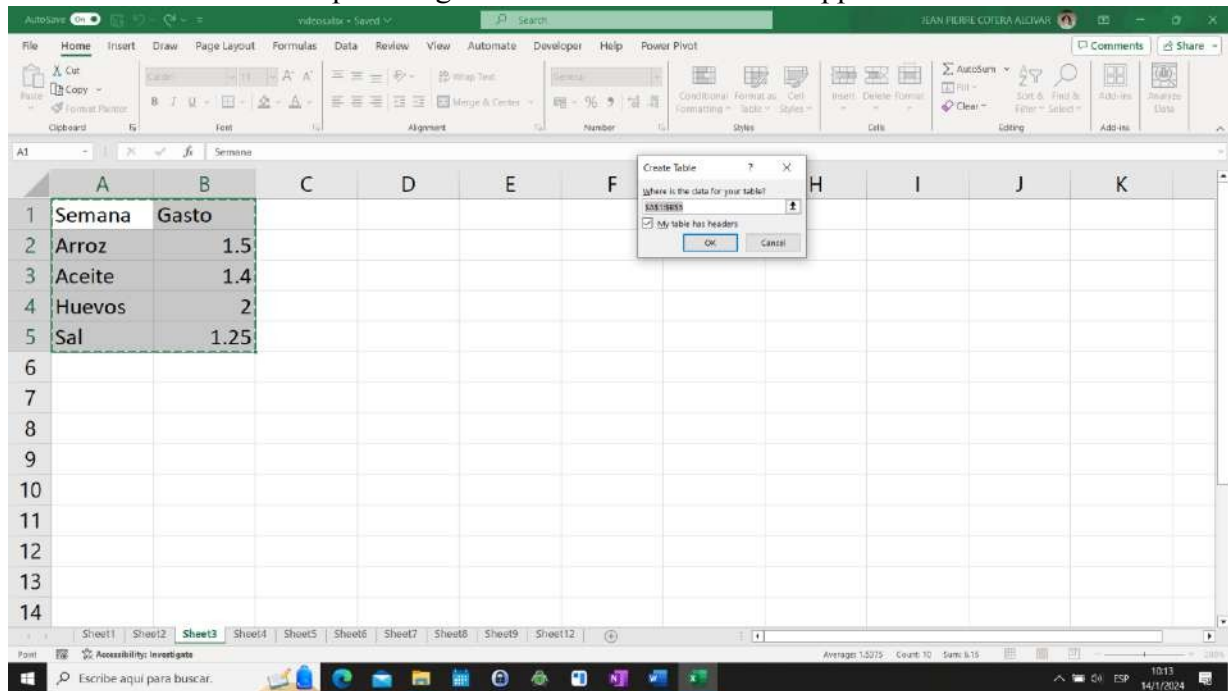
c. Then we go to the home tab and the table format option.



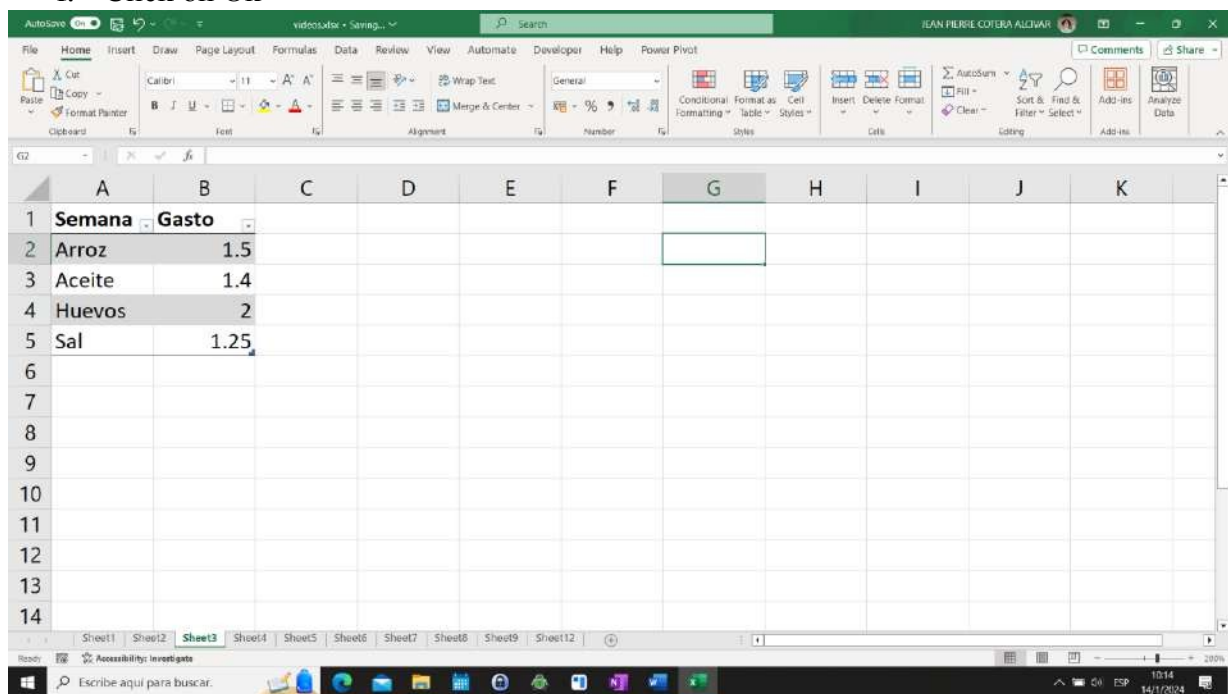
d. Clicking on it will open a box



e. Choose the first option Light and a small window will appear.

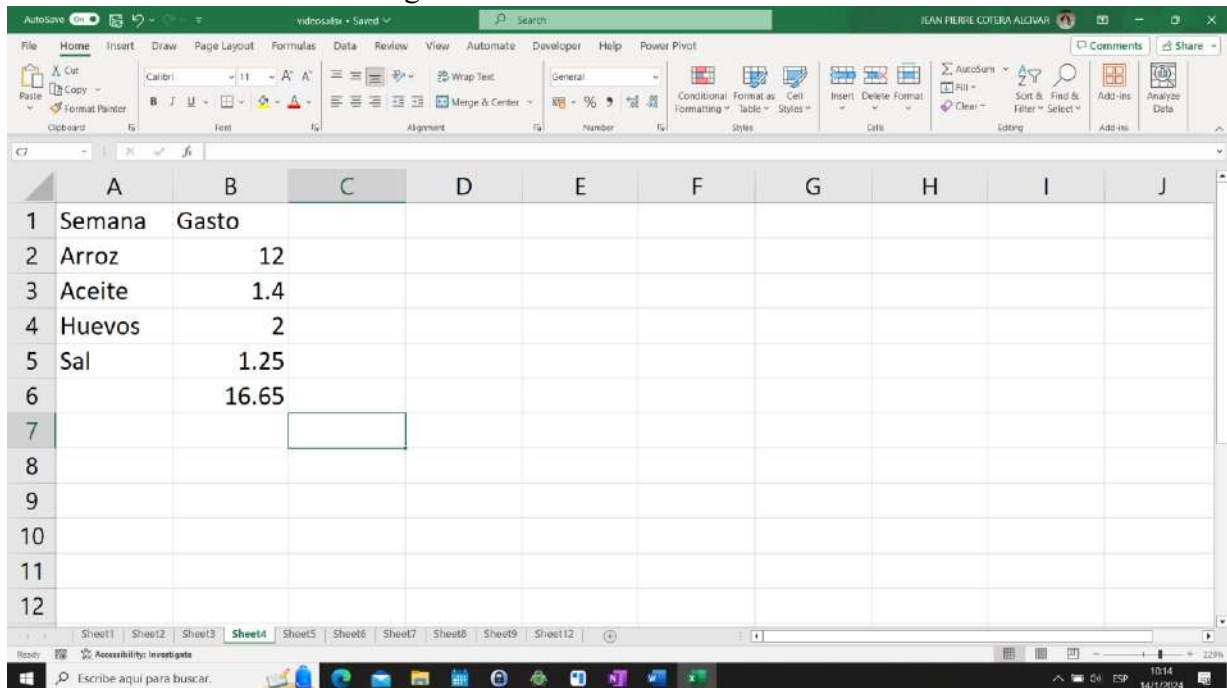


f. Click on Ok



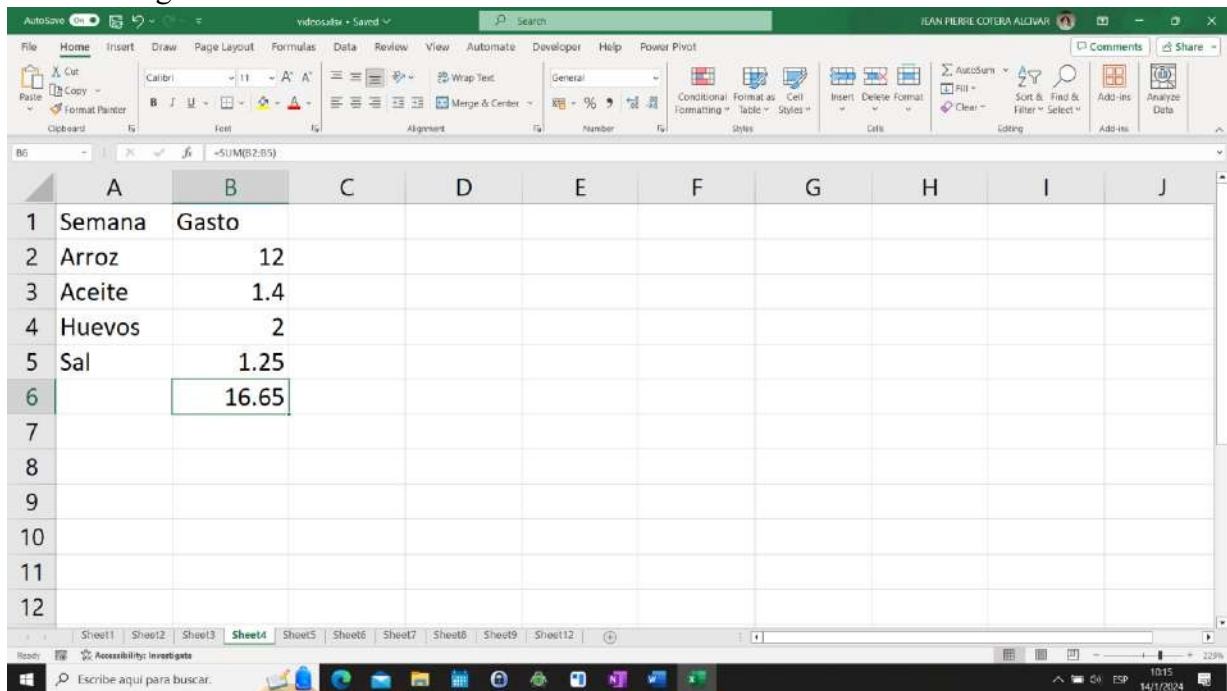
Color cell with conditions

a. We have the following data



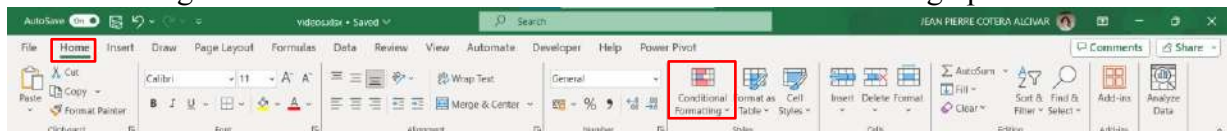
	A	B	C	D	E	F	G	H	I	J
1	Semana	Gasto								
2	Arroz	12								
3	Aceite	1.4								
4	Huevos	2								
5	Sal	1.25								
6		16.65								
7										
8										
9										
10										
11										
12										

b. We go to cell B6 which has the total sum.

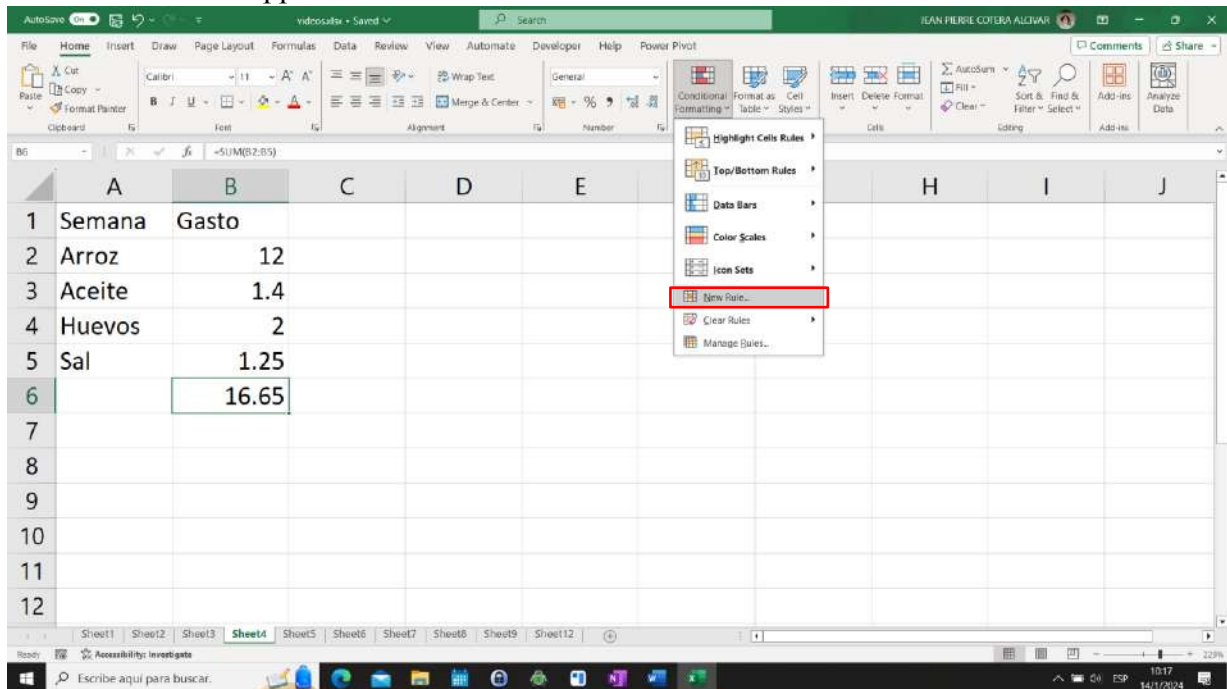


	A	B	C	D	E	F	G	H	I	J
1	Semana	Gasto								
2	Arroz	12								
3	Aceite	1.4								
4	Huevos	2								
5	Sal	1.25								
6		16.65								
7										
8										
9										
10										
11										
12										

c. Then go to the home tab and click on the Conditional Formatting option.



d. A box will appear and we will choose New rule.



e. In the minimum part we will do the following. In type we will place Number and then we will put 0 (zero). In the maximum in type we will place Number (number) and then we will put 40 (forty).

New Formatting Rule

Select a Rule Type:

- Format all cells based on their values
- Format only cells that contain
- Format only top or bottom ranked values
- Format only values that are above or below average
- Format only unique or duplicate values
- Use a formula to determine which cells to format

Edit the Rule Description:

Format all cells based on their values:

Format Style: 2-Color Scale

Minimum

Type: Number

Value: 0

Color: [Orange]

Maximum

Type: Number

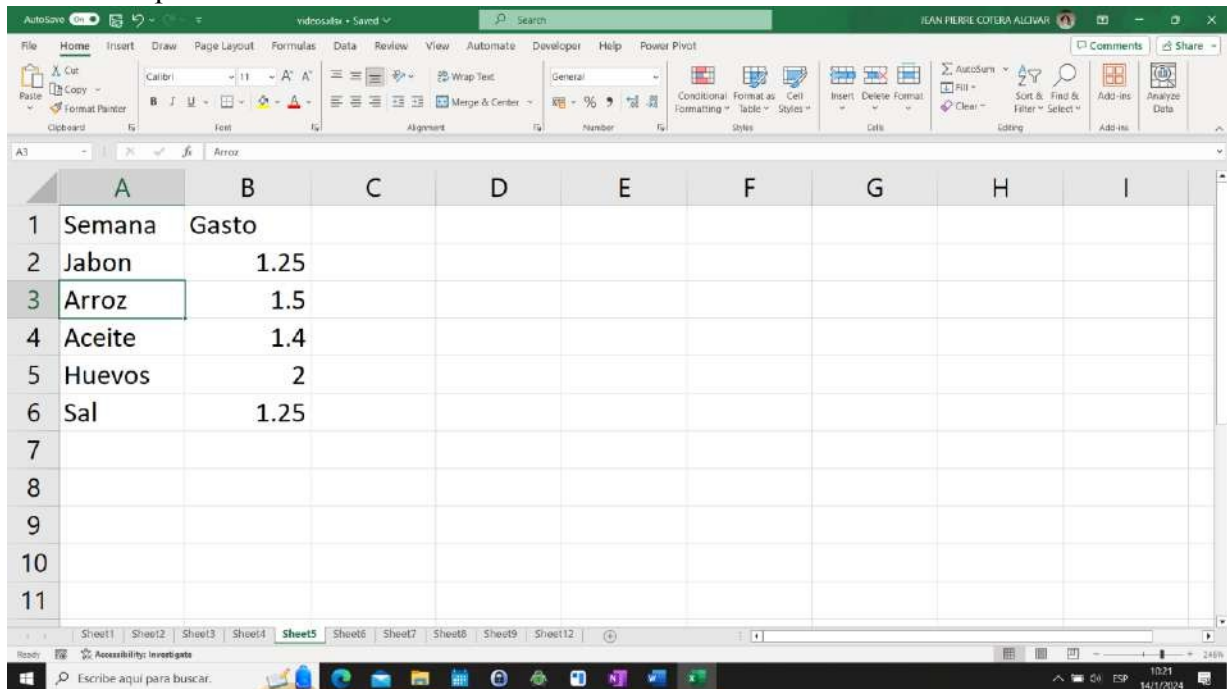
Value: 40

Color: [Yellow]

Preview: [Orange to Yellow gradient bar]

OK Cancel

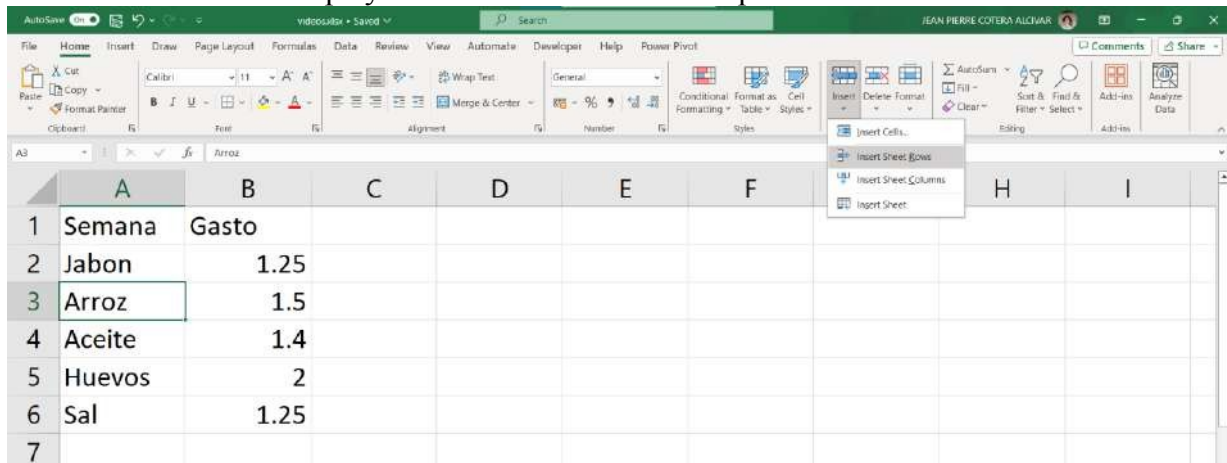
b. We place ourselves in the Rice cell



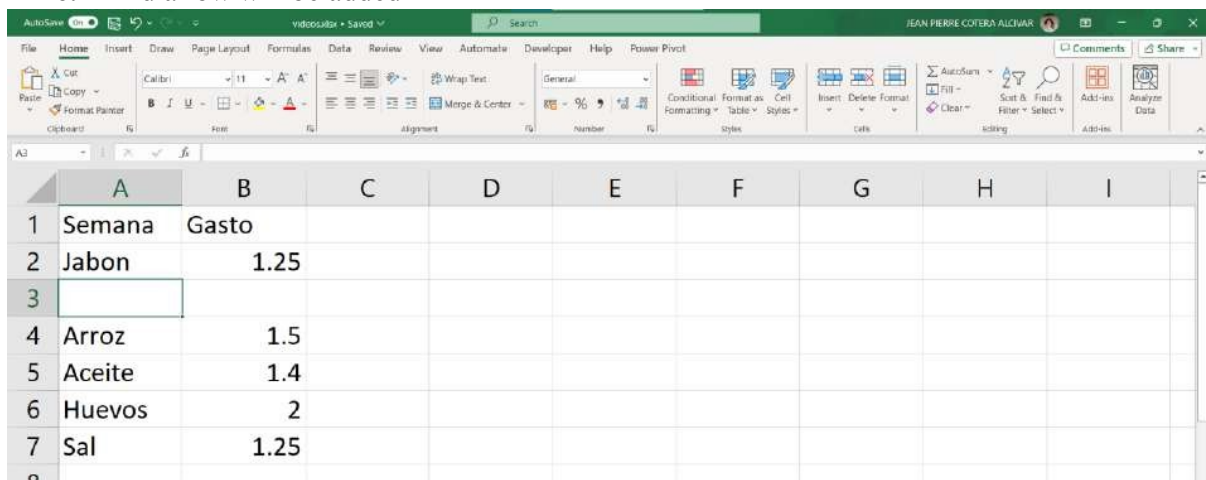
c. Then we go to the Home tab and then to the insert option



d. A box will be displayed and we choose the row option

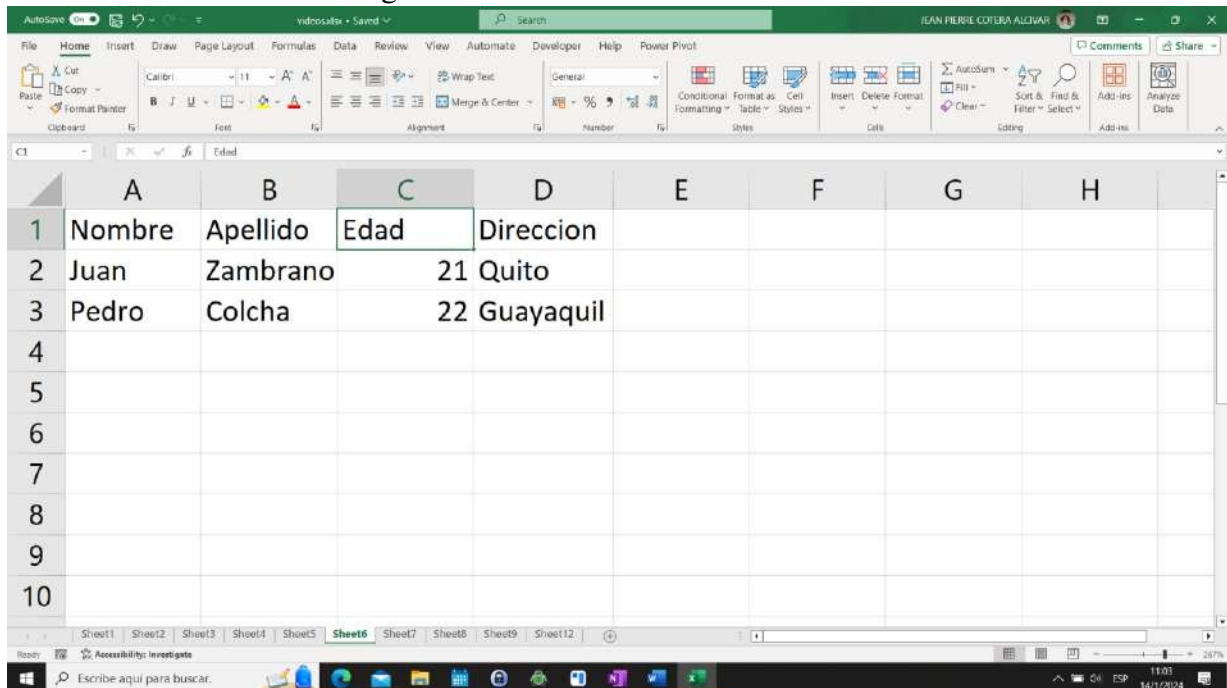


e. And a row will be added



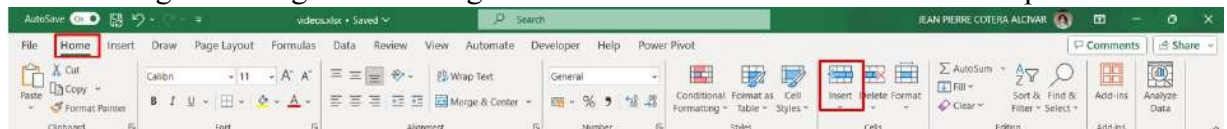
Add a column

a. We have the following data

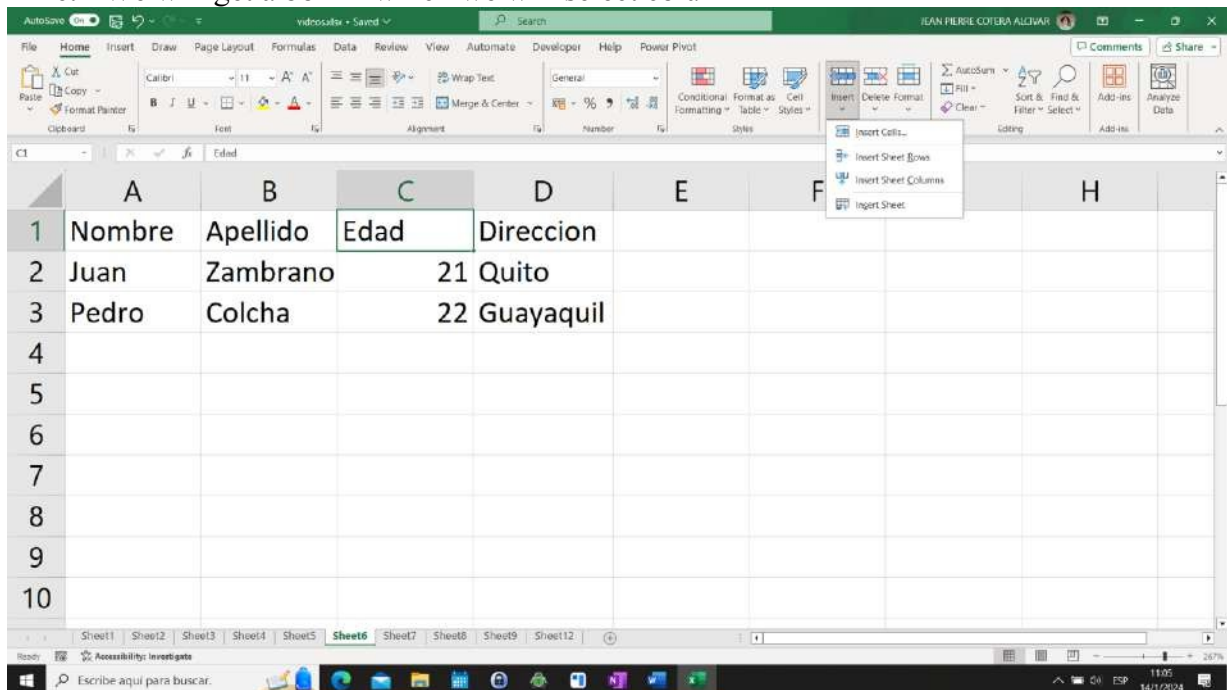


	A	B	C	D	E	F	G	H
1	Nombre	Apellido	Edad	Direccion				
2	Juan	Zambrano	21	Quito				
3	Pedro	Colcha	22	Guayaquil				
4								
5								
6								
7								
8								
9								
10								

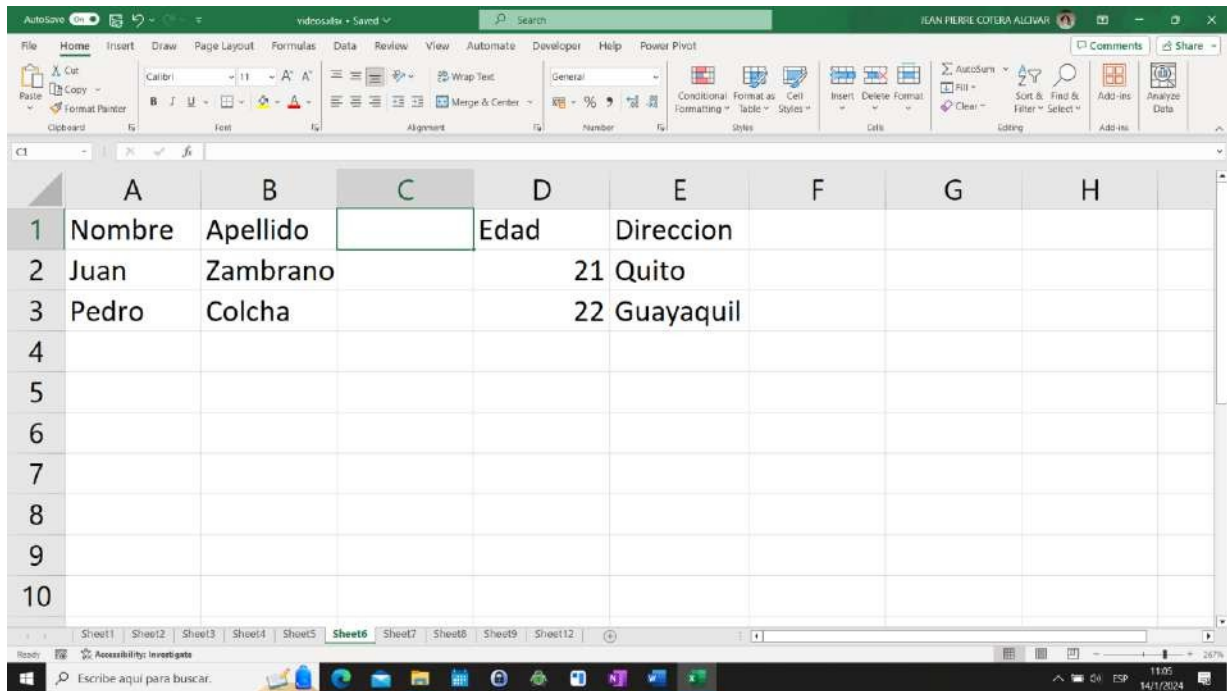
b. We go to the age cell. Then go to the home tab and click on the insert option.



c. We will get a box in which we will select column

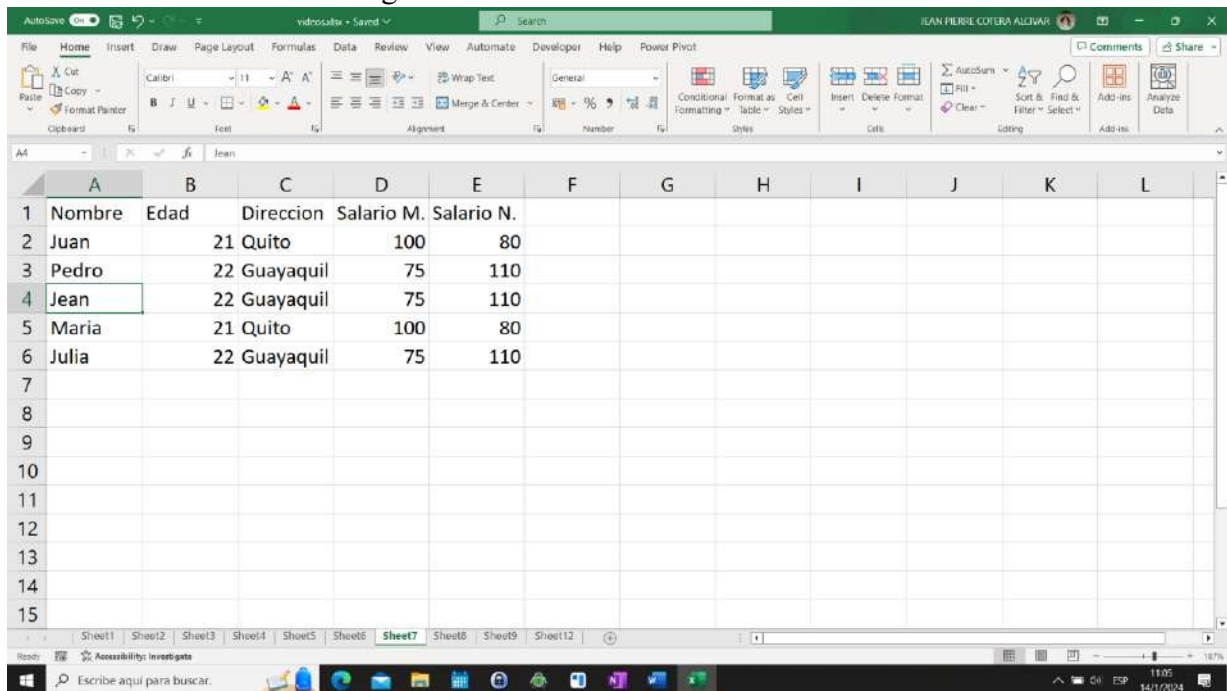


d. And a new column will be added



Delete a row

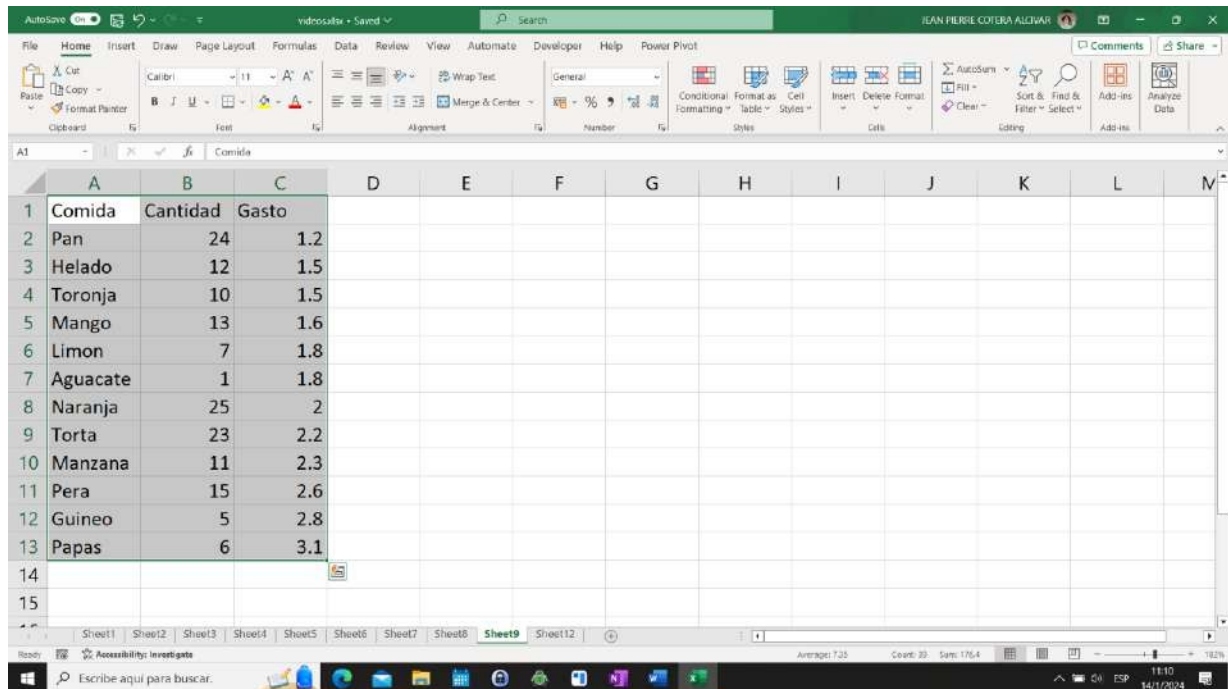
a. We have the following data



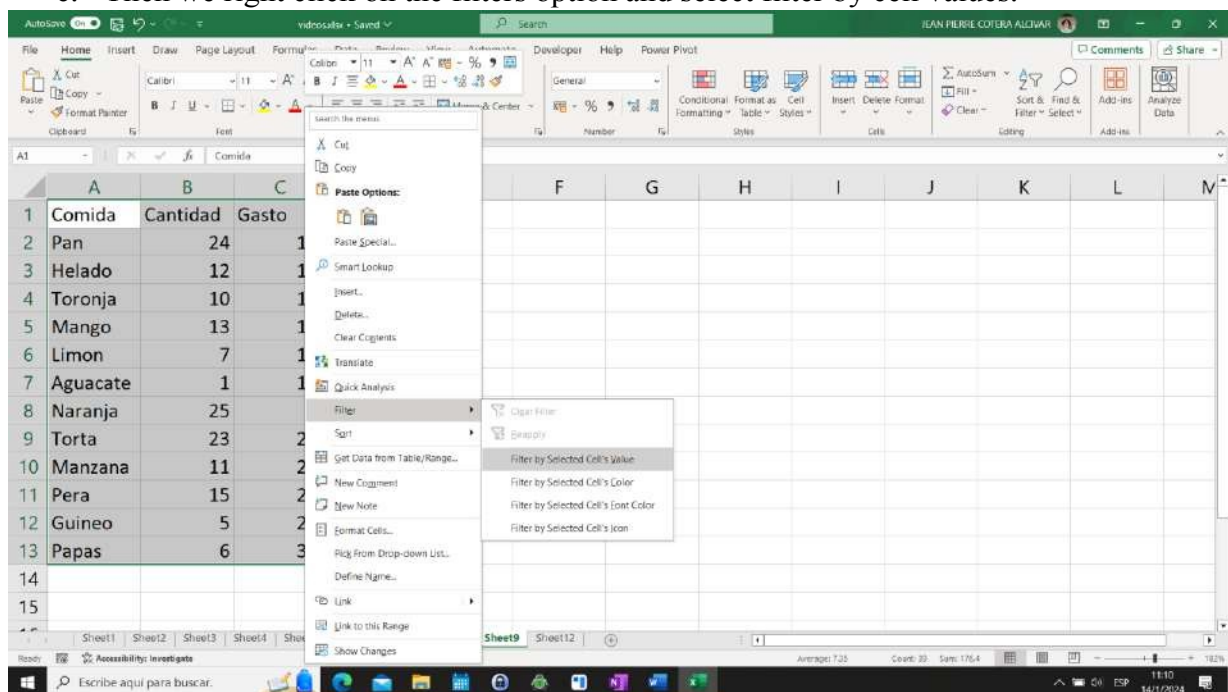
b. We go to the Jean cell. Then go to the Home tab and click on the delete option.



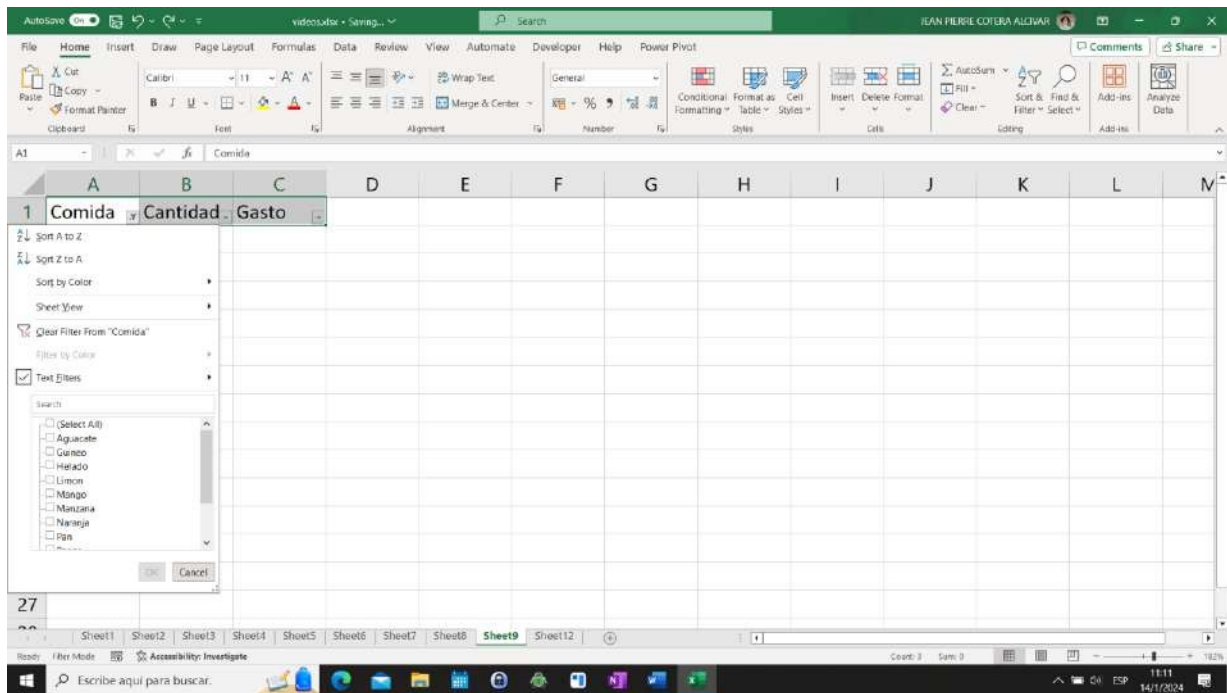
b. Select the entire table with its headers



c. Then we right click on the filters option and select filter by cell values.



d. Then click on the little arrow in the lower corner of the header to see the values.



e. Select All and click on Ok

AutoSave | WordSource | Saved | Search | JEAN PIERRE COTERA ALVARO | Comments | Share

File Home Insert Draw Page Layout Formulas Data Review View Automate Developer Help Power Pivot

Clipboard Font Styles Alignment Number

Calibri 11 A A' Bold Italic Underline Text Color Background Color Merge & Center Wrap Text

General Conditional Formatting Cell Styles

Insert Delete Format

Autosum Fill Sort & Filter Find & Select Add-ins Analyze Data

A14

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Comida	Cantidad	Gasto										
2	Pan	24	1.2										
3	Helado	12	1.5										
4	Toronja	10	1.5										
5	Mango	13	1.6										
6	Limon	7	1.8										
7	Aguacate	1	1.8										
8	Naranja	25	2										
9	Torta	23	2.2										
10	Manzana	11	2.3										
11	Pera	15	2.6										
12	Guineo	5	2.8										
13	Papas	6	3.1										
14													
15													

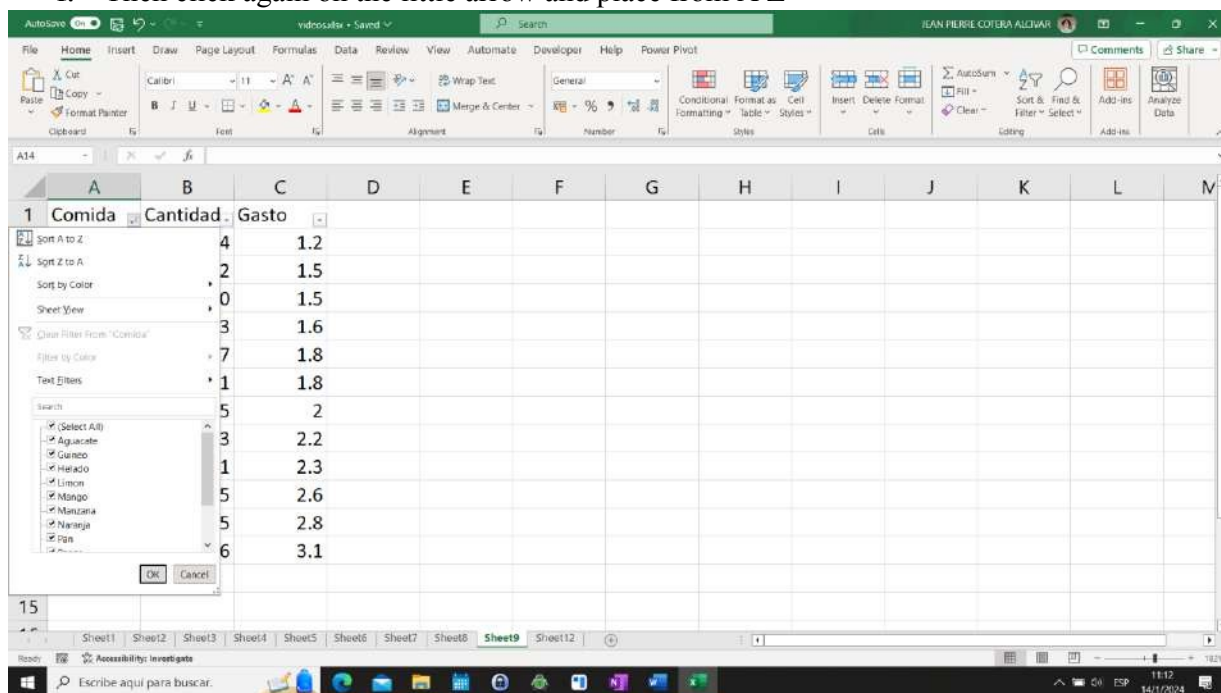
Sheet1 Sheet2 Sheet3 Sheet4 Sheet5 Sheet6 Sheet7 Sheet8 Sheet9 Sheet12

Ready Accessibility: Insert Paste

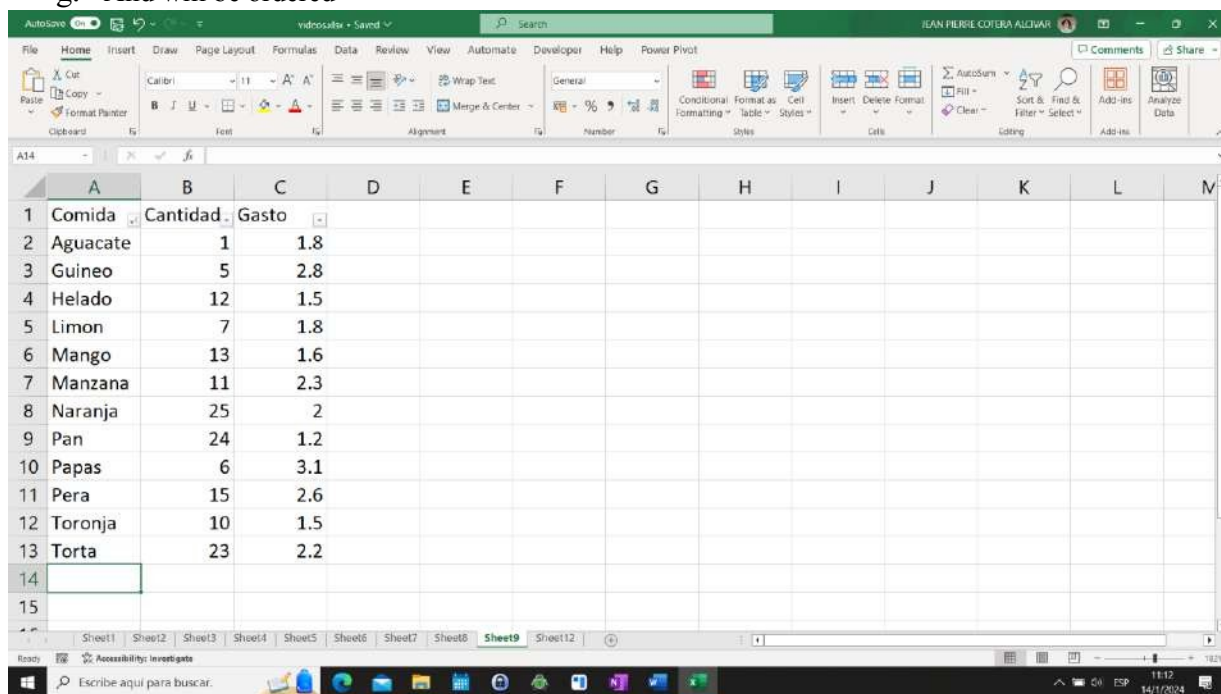
Escribe aquí para buscar.

11:12 14/1/2024

f. Then click again on the little arrow and place from A-Z

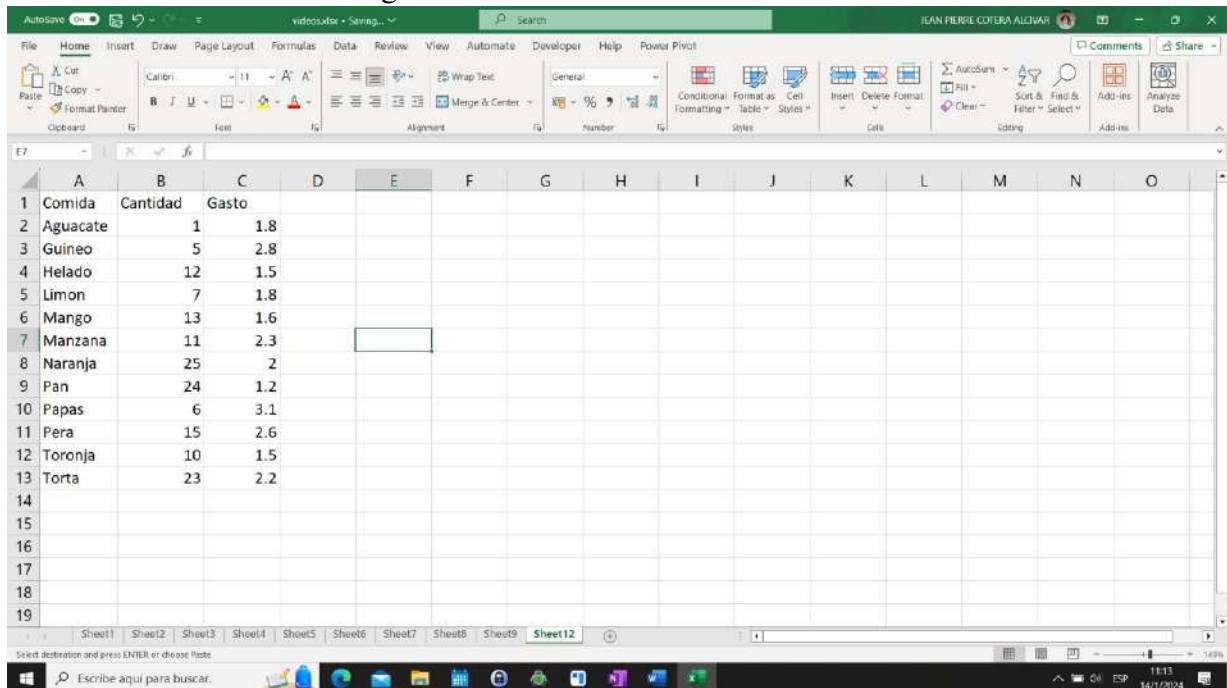


g. And will be ordered



Export spreadsheet as PDF

a. We have the following data



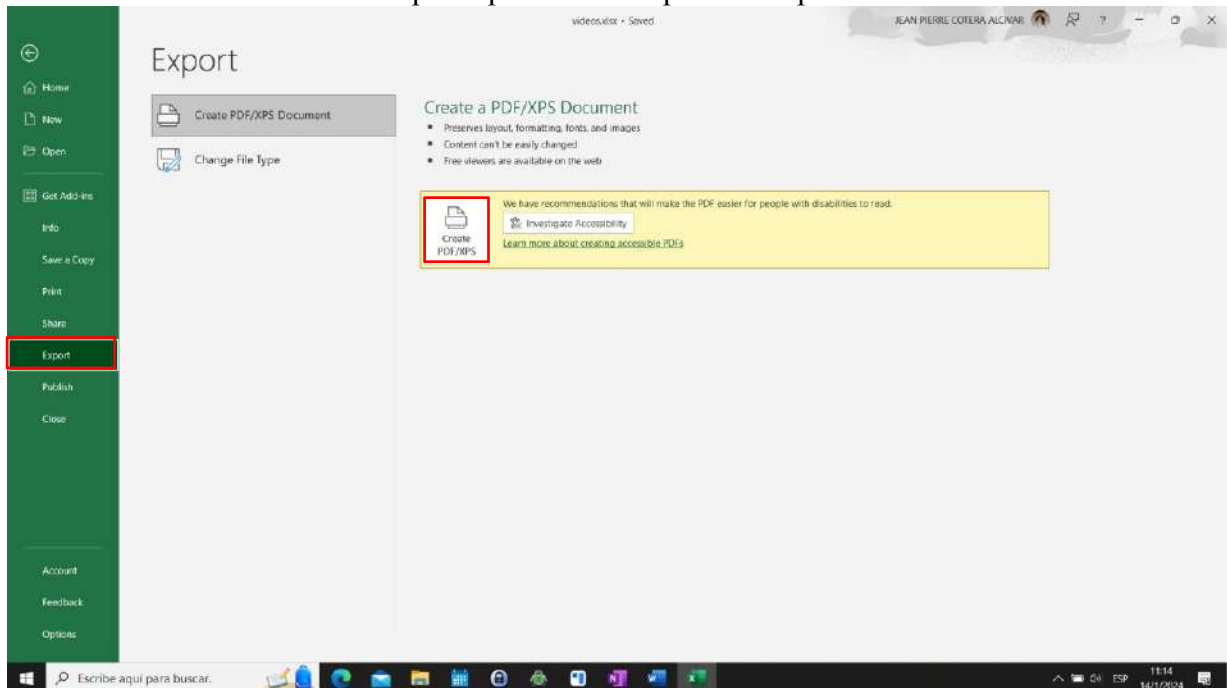
The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Comida	Cantidad	Gasto												
2	Aguacate	1	1.8												
3	Guineo	5	2.8												
4	Helado	12	1.5												
5	Limon	7	1.8												
6	Mango	13	1.6												
7	Manzana	11	2.3												
8	Naranja	25	2												
9	Pan	24	1.2												
10	Papas	6	3.1												
11	Pera	15	2.6												
12	Toronja	10	1.5												
13	Torta	23	2.2												
14															
15															
16															
17															
18															
19															

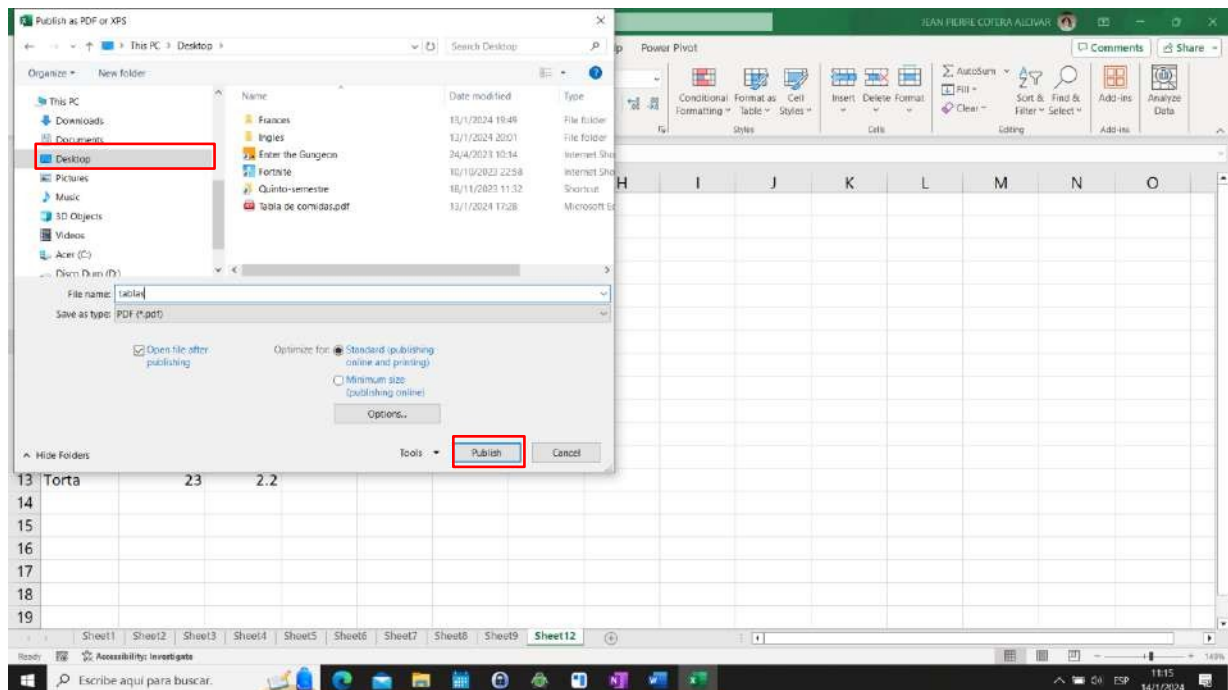
b. We go to the file tab



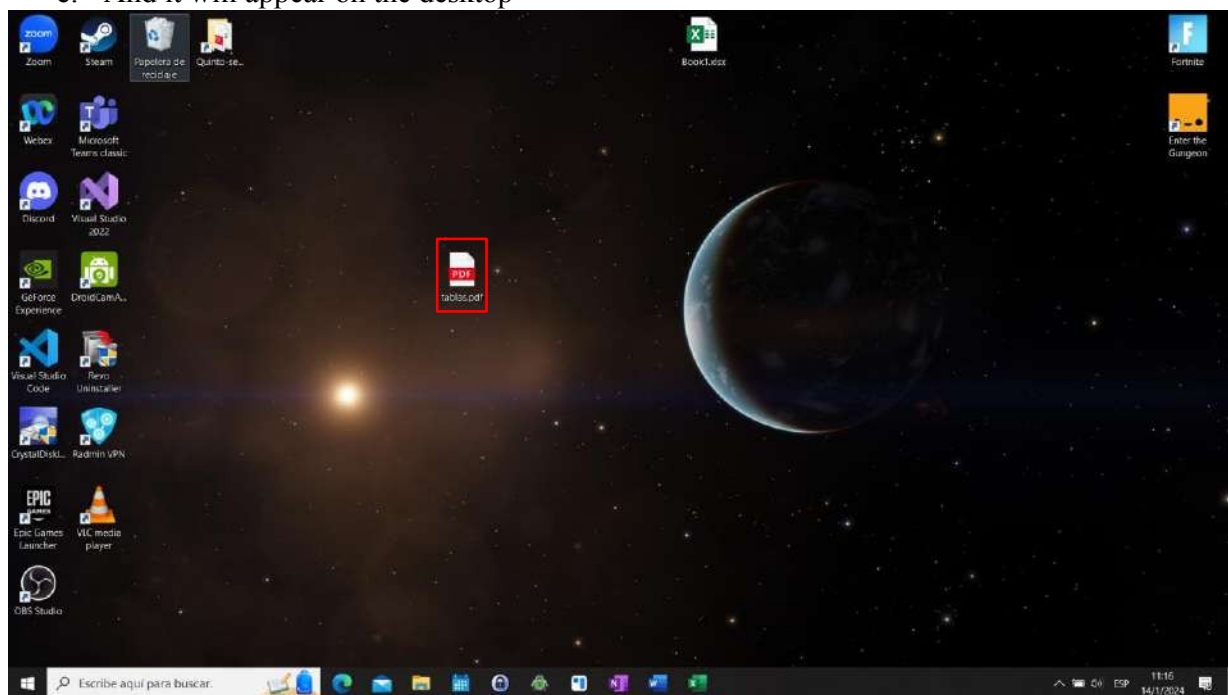
c. Then we click on the export option and we put create pdf



- d. We get a box to save in some place that pdf. We go to desktop and we name it tables. Then click on publish



- e. And it will appear on the desktop



8. Frequently Asked Questions

8.1. How can I change the formatting of a cell in Excel?

To change the formatting of a cell, select the cell or range, go to the "Home" tab, and use the options in the "Format Cells" group to adjust the numeric, date, or text formatting, among others.

8.2. Is it possible to undo multiple actions in Excel?

Yes, you can undo several actions in Excel. Use the "Undo" button on the quick access bar or the keyboard shortcut (Ctrl + Z) several times to reverse the actions performed.

8.3. How can I perform basic mathematical operations in a cell?

To perform basic mathematical operations, enter the formula directly in the formula bar. For example, to add, you can type "`=A1+B1`" to add cells A1 and B1.

8.4. What is the function of the "Find" function in Excel?

The "Find" function allows you to search for a specific value in a spreadsheet. You can access it from the "Home" tab and use it to quickly locate data.

8.5. How can I protect a spreadsheet in Excel?

To protect a spreadsheet, go to the "Review" tab and use the "Protect sheet" option. You can set a password and select the allowed actions, such as editing cells or inserting rows.

8.6. What is the difference between a formula and a function in Excel?

A formula is a mathematical expression that performs calculations, while a function is a predefined formula that performs a specific operation. For example, "`=SUM(A1:A5)`" is a formula that sums the range A1 to A5, while "`=AVERAGE(B1:B10)`" is a function that calculates the average of the range B1 to B10.

9. Glossary

Term	Description
Cell	The basic unit in an Excel spreadsheet. A cell is identified by its location at the intersection of a row and a column and can contain data, formulas or functions.
Conditional Formatting	An Excel function that allows you to apply visual formatting to a range of cells based on specific conditions. Facilitates quick identification of patterns and trends in the data.
Formula:	A mathematical or logical expression used to perform calculations in Excel. Formulas can involve operators, cell references and predefined functions.
Formula Bar:	The section at the top of the Excel window that displays the formula or the contents of the active cell. It is the place where data and formulas are entered or modified.
Spreadsheet:	An individual page within an Excel workbook. Each spreadsheet can contain data, formulas, graphs and other elements related to a specific set of information.
AutoSum:	A function that automatically performs the sum of a selected range of cells. It makes it easy to quickly perform simple mathematical operations without the need to manually enter the sum formula.

10. Bibliography and References

Reference	Title
[1]	Gudino, J. E. (2020). Excel basico: Guia practica. Independently Published.